

# **Telco Customer Churn Prediction & Retention Strategy Case Study**

**An End-to-End Machine Learning & Power BI Solution**

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**Tools Used:**

**Python – Scikit-Learn – Pandas – Power BI**

# **Table of Contents**

- 1. Executive Overview**
- 2. Business Problem**
- 3. Data Summary**
- 4. Exploratory Data Analysis**
- 5. Modeling Summary**
- 6. Churn Drivers**
- 7. Unified Dataset for Power BI**
- 8. Power BI Dashboard Overview**
- 9. Strategic Recommendations**
- 10.Customer Personas**
- 11.Expected Business Impact**
- 12.Conclusion**

### 1. Executive Overview

Customer churn remains one of the most costly challenges in the telecommunications industry. Retaining existing customers is significantly more cost-effective than acquiring new ones, making churn prediction and targeted retention strategies essential for operational success.

This project delivers a **complete, end-to-end churn management framework** that integrates:

- **Python-based data cleaning, feature engineering, and machine learning**
- **Exploratory data analysis to uncover churn drivers**
- **A unified dataset for BI consumption**
- **A multi-page Power BI dashboard**
- **Strategic retention recommendations**
- **Human-readable customer personas**

The result is a scalable, interpretable, and business-ready solution that identifies at-risk customers, explains why they churn, and provides actionable strategies to reduce churn by an estimated **10–15%**.

### 2. Business Problem

Telecom providers face churn driven by:

- Pricing pressure
- Service dissatisfaction
- Contract flexibility
- Payment method friction
- Fiber optic performance expectations

Without a systematic approach to identifying at-risk customers, retention teams struggle to:

- Prioritize outreach
- Allocate incentives effectively
- Understand customer behavior
- Reduce churn at scale

**Objective:** Predict customer churn, understand its drivers, segment customers by risk, and deliver a Power BI dashboard that enables targeted retention strategies.

### 3. Data Summary

The dataset includes **7,032 customers** and **21 features** covering:

- Demographics
- Account information
- Service subscriptions
- Billing details

#### Feature Engineering Included:

- Tenure buckets
- One-hot encoded categorical variables
- Scaled numeric features
- Preserved index for merging predictions
- Dashboard-ready risk segmentation (Low, Medium, High, Critical)

These engineered features formed the foundation for both modeling and BI visualization.

### 4. Exploratory Data Analysis (EDA) — Key Insights

EDA revealed several strong churn patterns:

- **Low tenure customers churn at the highest rate**
- **Month-to-month contracts show significantly higher churn**
- **Electronic check users churn more than automatic payment users**
- **Fiber optic customers churn more than DSL customers**
- **Higher monthly charges correlate with increased churn**

These insights guided feature engineering, model selection, and dashboard design.

## 5. Modeling Summary

Three models were trained and evaluated:

| Model               | Accuracy | Recall (Churn) | ROC-AUC |
|---------------------|----------|----------------|---------|
| Logistic Regression | 0.80     | 0.57           | 0.836   |
| Random Forest       | 0.80     | 0.60           | 0.824   |
| XGBoost             | 0.78     | 0.53           | 0.820   |

**Selected Model: Logistic Regression**

Chosen for:

- Strong ROC-AUC
- Interpretability
- Stable probability outputs
- Ideal behavior for risk segmentation

### Top Churn Drivers (Feature Importance)

1. TotalCharges
2. Tenure
3. MonthlyCharges
4. Contract: Month-to-Month
5. OnlineSecurity: No
6. TechSupport: No
7. InternetService: Fiber Optic
8. PaymentMethod: Electronic Check

These drivers align with EDA findings and provide clear business direction.

### 6. Unified Dataset for Power BI

A dashboard-ready dataset was created containing:

- customerID
- Actual churn label
- Predicted churn probability
- Risk segment (Low, Medium, High, Critical)
- Tenure group
- Monthly and total charges
- Encoded contract, payment, and service attributes

This dataset enables customer-level analysis, segmentation, and strategic targeting.

### 7. Power BI Dashboard Overview

The dashboard consists of **five pages**, each designed for business decision-making.

#### Page 1 — Churn Overview

- High-level KPIs
- Churn distribution
- Customer counts
- Quick segmentation filters

#### Page 2 — Risk Segmentation

- Breakdown of customers by churn probability
- High/Critical risk spotlight
- Interactive slicers

#### Page 3 — Churn Driver Insights

- Contract type
- Payment method
- Internet service

- Monthly charges
- Tenure buckets

### Page 4 — Retention Strategy

- Priority segments
- Recommended actions
- Retention playbook
- Expected impact (10–15% churn reduction)

### Page 5 — Customer Personas

Human-readable profiles of at-risk segments:

- **Price-Sensitive Sprinter**
- **Overloaded Fiber User**
- **Low-Engagement Legacy Customer**

These personas translate analytics into real customer behavior.

## 8. Strategic Recommendations

Based on the combined Python + Power BI analysis:

### 1. Strengthen onboarding for new customers

Low tenure is the strongest churn driver.

### 2. Promote long-term contracts

Month-to-month customers churn at the highest rate.

### 3. Encourage automatic payment methods

Electronic check users churn significantly more.

### 4. Investigate fiber optic service issues

Fiber optic customers show elevated churn.

### 5. Offer pricing relief for high-charge customers

Monthly charges strongly correlate with churn.

### **6. Increase adoption of support/security services**

Lack of OnlineSecurity and TechSupport increases churn.

### **7. Prioritize outreach to High and Critical risk segments**

These segments represent the highest ROI for retention efforts.

## **9. Expected Business Impact**

Implementing the recommended retention strategies is expected to:

- Reduce churn among high-risk customers by **10–15%**
- Improve customer satisfaction
- Increase contract stability
- Reduce support costs
- Improve lifetime value (LTV)

This impact is based on model predictions, segment sizes, and industry benchmarks.



### 10. Conclusion

This project delivers a **complete churn prediction and retention strategy solution** that integrates:

- Machine learning
- Exploratory analytics
- Business intelligence
- Strategic recommendations
- Customer personas

The result is a **scalable, interpretable, and business-ready churn management framework** that empowers telecom providers to:

- Identify at-risk customers
- Understand churn drivers
- Deploy targeted retention strategies
- Improve customer satisfaction
- Reduce churn at scale